

Introduction:

Machine Learning Engineer with a Ph.D. in Particle Physics and over 5 years of experience in leveraging AI to tackle complex problems. My journey, marked by overcoming visual challenges, has equipped me with a distinctive perspective in AI and computer vision. I specialize in translating theoretical concepts into high-impact solutions, seamlessly integrating cutting-edge ML technologies with traditional mathematical techniques. My expertise spans computer vision, natural language processing, and AI-driven systems, driven by a commitment to innovation and practical application.

Skills:

- **Languages:** Python, C, C++, Rust, SQL.
- **Tools:** Docker, Git.
- **Libraries/Frameworks:** OpenCV, PyTorch, TensorFlow, Huggingface, Torch.

Experience

CV Engineer

Cloudphysician

2nd Dec 2024 - Present

I contribute to Cloudphysician, a mature startup innovating at the intersection of AI and medicine, by building ML-powered solutions that transform healthcare and intensive care. My work involves developing algorithms that process multimodal data (text, videos, images, and audio) to automate laborious processes and provide crucial decision support, ultimately improving patient care.

Respiratory rate:

- Engineered a non-contact video-monitoring system to predict neonatal respiration rates by leveraging **deep compression autoencoders** (spatial) and Fourier transforms (temporal) for feature extraction.
- Reduced the burden of manual NICU respiratory checks, saving one minute per baby of nursing time per hour,

Fall risk:

- Engineered a multi-head image classification model with the **EVA** backbone to predict ICU patient fall risks.
- Projected to increase monitoring capacity by 4x (from 6 to 24 patients per supervisor).

Also involved in other image and video classification efforts to

- identify respiratory distress among severely sick patients.
- Identify and track various objects present in the ICU.
- Developed code to translate the image coordinates (X, Y) and bounding boxes to camera coordinates (pan, tilt, zoom).

ML Engineer

TRI3D, Alphabake

16th Jan 2023 - 27th Nov 2024

At this innovative early-stage company, I drove innovation at the intersection of AI and fashion technology. Through innovative AI implementations and workflow optimizations, I spearheaded a dramatic 97% reduction in manual image editing time, slashing the average effort from 2 hours to just 4 minutes per image. This breakthrough significantly enhanced team productivity and enabled a 30-fold increase in daily image processing capacity.

Responsibilities:

- Develop and deploy cutting-edge computer vision and image generation solutions for fashion applications, including:
 - Virtual try-on systems
 - Background/model swapping
 - Mannequin-to-model image conversion
- Design scalable architecture for AI-driven image processing systems
- Enhance image editing tools and workflows using machine learning
- Conduct research to adapt and implement state-of-the-art computer vision techniques for the fashion industry
- Oversee ML model deployment and finetuning in production environments, collaborating with image editors for seamless integration
- Leveraging cutting-edge technologies, including large language models and multimodal vision-language models, to streamline our workflow, automate repetitive tasks (like quality control), and enhance the preparation of training data.
- Integrated traditional CV techniques (like fourier transforms, histogram reshaping, ...) with modern methods for efficient image processing and refinement.

Tech Stack:

- **Core Technologies:** OpenCV, PyTorch, Nginx, Flask, FastAPI
- **Semantic Segmentation:** MMSegmentation, cloth-segmentation, SAM, transparent-background, MVANet, SCHP
 - Enhanced MVANet training with **Prodigy** optimizer and robust image augmentation using **Albumentations** and **Kornia**
- **AI Libraries and Tools:**
 - **Hugging Face** ecosystem: diffusers, transformers, accelerate, safetensors, datasets, PEFT
 - Diffusion fine-tuning: **x-flux**, **sd-scripts**, **EveryDream2trainer**
 - **IP-Adapters:** Trained with private data and enhancements
 - Automated image processing: **lmdeploy**, **ollama** for captioning, filtering, comparison, and prompt enhancements
- **Workflow Development:** ComfyUI ecosystem and plugins for collaborative workflow development with image editors
 - Contributed to multiple open-source ComfyUI plugins

Applied Scientist - II (L5)

Alexa - ASR, Amazon

27th June 2022 - 28th December 2022

In this role, my main responsibility is to improve the accuracy of automatic speech recognition (ASR) for alexa.

1. AM-LM Fusion:

- Incorporated techniques from **2201.10240** to improve fusion of acoustic and text representation.
- This resulted in 1% word error rate improvement over the baseline conformer model.

3. Neural biasing for rare words:

- Implemented **Neural biasing** to improve identification of rare words such as contact names.

2. HuBERT pre-trained conformer teacher:

- Implemented techniques from **HuBERT** pre-training with a large conformer model.
- Resulted in around 10% improvement in word error rate over the existing baseline for the labeling of SSL data.

4. Speaker adaptation:

- Implemented speaker adaptation to improve ASR accuracy.

Research Scientist - II (L5)

Lastmile, Amazon

4th May 2020 - 24th June 2022

In this role, my main responsibility was geocoding (translating human readable address to GPS coordinates).

1. Checking validity of outlier deliveries:

- **Goal:** Find cause for disagreement between location vended by Amazon geocoding systems and location of the actual delivery.
- Developed a supervised ML model based on boosted decision trees
- This led to an improvement in F1 score from 0.71 (previous model based on heuristics) to 0.88 and expected to contribute around 20% improvements to the actionability during manual audit queues.

3. Geocoding with pair-wise ranking:

- **Geocoding** Translating human readable addresses into GPS coordinates.
- Worked on supervised pair-wise ranking model based on random forests.
- I was responsible for analyzing this and push it towards production for geocoding in France.
- This led to 1% improvement in mean geocoding accuracy, 10% improvement in coverage and significant drop in deliveries to wrong locations and unlocatable address.

2. Analyzing parking locations:

- **Goal:** To find good parking locations in busy cities before making the delivery.
- Provide many improvements to the existing approach using an ML model which classified delivery person activities.
- Led to 2% improvement in mean service times over the previous model.

4. Geocoding for new addresses:

- Geocoding for new addresses (addresses with no historic delivery data) is a challenging task.
- Developed an NLP (seq to seq) transformer model mapping known addresses to geocodes.
- Developed and validated a linear interpolation model to predict coordinates of unknown addresses.

Part time / Intern

Cloudphysician

1st Jan 2020 - 1st March 2020

Identify ventilator images, digitize pressure and volume flow graphs, classify disease state.

Open source contributions

Linux kernel audio driver regression

Bug 219897

- Reported an audio driver regression that happened in the Linux kernel.
- Identified the code and commit which caused the regression.
- Contributed to debugging and fixing the issue.

Contributed to various image generation related projects

ComfyUI Plugin Eco System

- Contributed to fixing a tensor shape mismatch bug in **Tiled Diffusion** plugin (this fix was incorporated into the parent repo).
- Contributed to fixing a bug in **ComfyUI workflow to python extension** which prevented it from working with newer versions of **ComfyUI**.
- Implemented a virtual tryon approach using LORA and Controlnets in flux using the **x-flux** library.
- Contributed numerous ComfyUI plugins to **tri3d-comfyui-nodes**.
- Contributed to fixing a shape mismatch issue in **ComfyUI IPAdapter plus** plugin.

Education

PhD (Particle Physics)

TIFR, Mumbai

August 2014 - February 2020

Ph.D in High Energy Phenomenology with expertise in advanced math, high performance computing, data analysis and machine learning.

Int. BS, MS (Physics, Math)

IISER, Pune

August 2009 - July 2014 (CGPA: 8.1)

Bachelor and master of science in the field of physics and mathematics.

Research Projects

The research topic of my PhD was in the field of phenomenological particle physics covering topics like Beyond the Standard Model (SM) of Particle Physics and application of machine learning for QCD jet substructures.

Publications

1. **Boosted top quark tagging and polarization measurement using machine learning:**
Soham Bhattacharya, Monoranjan Guchait, **Aravind H. Vijay**.
[Phys. Rev. D **105**, 042005](#)
([arXiv:2010.11778 \[hep-ph\]](#)).
2. **A robust anomaly finder based on autoencoder:**
Tuhin S. Roy, **Aravind H. Vijay**.
([arXiv:1903.02032 \[hep-ph\]](#)).
3. **Boosted Top quark polarization:**
Rohini Godbole, Monoranjan Guchait, Charanjit K. Khosa, Jayita Lahiri, Seema Sharma, **Aravind H. Vijay**.
[Phys. Rev. D **100**, 056010](#)
([arXiv:1902.08096 \[hep-ph\]](#)).
4. **Probing Heavy Charged Higgs Boson at the LHC:**
Monoranjan Guchait, **Aravind H. Vijay**.
[Phys. Rev. D **98**, 115028 \(2018\)](#)
([arXiv:1806.01317 \[hep-ph\]](#)).
5. **Detection prospects of light pseudoscalar Higgs boson at the LHC:**
Monoranjan Guchait, **Aravind H. Vijay**, Jacky Kumar.
[J. High Energ. Phys. \(2017\) 2017: 122](#)
([arXiv:1705.06275 \[hep-ph\]](#)).
6. **On the superposition principle in interference experiments:**
Aninda Sinha, **Aravind H. Vijay**, Urbasi Sinha.
[Sci. Rep. 5, 10304; doi: 10.1038/srep10304 \(2015\)](#).
([arXiv:1412.2198 \[quant-ph\]](#)).